



Improving Metadata and Discoverability

Making your software findable

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2026-09-23

Findability comes from description

A Zenodo record with a bare title and no keywords is a catalog record with no subject headings: retrievable if you already know it exists, invisible if you don't.

The tools are GitHub topics, Zenodo fields, and a README. The question is the same: *what would a seeker search for, and does this record contain it?*

You already have a license, a CITATION.cff, and a DOI. **This episode ties them together.**

What counts as useful metadata?

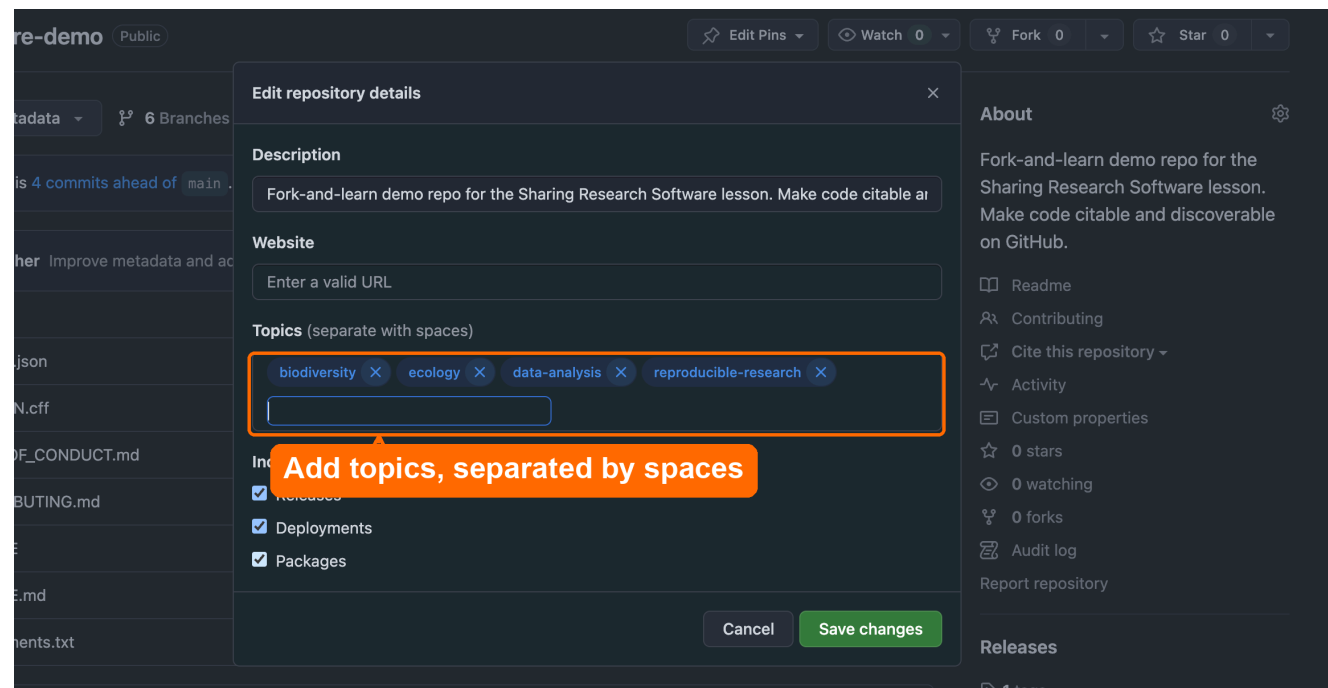
Good metadata answers predictable questions with minimal effort from the reader:

- **What is this software?** — short description or abstract
- **Who made it?** — authors, ORCIDs
- **How do I cite it?** — CITATION.cff + DOI
- **What domain is it for?** — keywords
- **What else does it relate to?** — related article, dataset, funding source

Add GitHub Topics

On your repo's front page, click the **gear icon next to About**, then fill in **Topics**:

- **Discipline tags** — `geospatial`, `text-mining`, `materials-science`
- **Method tags** — `simulation`, `visualization`, `machine-learning`
- **Language tags** — `python`, `r`



The 30-second rule

Your README is your software's front door.

If people can't tell what it does, how to install it, and how to use it in **30 seconds**, they leave.

A repo without a real README is a collection without a finding aid: only its author can use it.

README: 7 essential sections

1. **About** — what it does, in 2-3 sentences
2. **Features** — key capabilities
3. **Getting Started** — prerequisites + install
4. **Usage** — minimal working example
5. **Citation** — link to CITATION.cff / DOI
6. **License** — state terms explicitly
7. **Contact** — how to get help

[UC OSPO README Guide](#) (*UC-specific*)

Before → after

✗ Before (`main` , starting state)

```
1 # Biodiversity Analysis Toolkit
2
3 Analysis tools for biodiversity
4 research data.
```

One line of description. No install steps, no usage, no citation, no license, no one to contact.

✓ After (`05-add-metadata` , abridged)

```
1 # Biodiversity Analysis Toolkit
2 [DOI badge] [License badge]
3 Analysis tools for biodiversity... <!-- 1 About -->
4
5 ## Features <!-- 2 -->
6 - Citable, open source, documented
7
8 ## Getting Started <!-- 3 -->
9 Prerequisites: Python 3.9+
10 ### Installation & Usage <!-- 4 -->
11 pip install -r requirements.txt
12 python src/analysis.py
13
14 ## Citation <!-- 5 -->
15 Cite via CITATION.cff or the DOI badge.
16 ## License <!-- 6 -->
17 BSD 3-Clause. See LICENSE.
18 ## Contact <!-- 7 -->
19 Open an issue in this repository.
```

Beyond the README: community health files

CONTRIBUTING.md

How to contribute. Worth adding *if* you want contributions — or say plainly you don't.

CODE_OF_CONDUCT.md

Behavioral standards. [Contributor Covenant](#) is the standard choice.

CHANGELOG.md

Version history: what's new, what broke, what changed.

GitHub recognizes certain filenames

File	What GitHub does
README.md	Renders as the repository's landing page
LICENSE	Shows the license in the repo sidebar
CITATION.cff	Adds "Cite this repository" with APA/BibTeX
CODE_OF_CONDUCT.md	Counts toward the community profile
CONTRIBUTING.md	Links guidelines when someone opens an issue/PR
SECURITY.md	Becomes the security policy

These only appear once the file is on the default branch. GitHub changes the details, so teach the pattern: the filename is the signal.

Where should software live?

Zenodo is the recommended deposit location for software:

- Archives each GitHub release automatically
- Mints a DOI for every version
- Registers records with DataCite, which discovery services harvest
- Free, run by CERN, recognized by journals and funders

Dataverse and Dryad are good for **datasets**, but don't auto-integrate with GitHub releases the same way.

UC-specific: all UC campuses have Dryad. If you deposit data there, Dryad prompts a linked Zenodo code deposit automatically.

Zenodo metadata is what discovery systems see

Zenodo registers each record with **DataCite**. Search tools and discovery services pick it up from there, so the Zenodo record is what they show.

Add or refine: authors + ORCIDs, keywords, related works, funding, a readable description.

Caution: if a repo has a `.zenodo.json`, Zenodo uses *that* file, not `CITATION.cff`. Keep them in sync or you'll publish two different citation records for the same software.

Supporting others: priority order when metadata is thin

1. A readable **description** — the biggest single loss when it's empty
2. **Author ORCIDs** — every J. Chen on Zenodo is indistinguishable without one
3. **Keywords**, reused consistently across GitHub, CITATION.cff, Zenodo
4. **Related works** — link the software to its paper, dataset, grant

Challenge: improve your Zenodo record



Open your record on **Zenodo Sandbox** (sandbox.zenodo.org) from the release episode. Select **Edit**, and add:

- keywords
- author ORCIDs
- a real description
- *optional*: a related paper or funder, if you have one

Save and publish the updated record when done.

Challenge: which change makes it discoverable?

 3 min

A tool has been public on GitHub for a year: one-line README, no topics, a Zenodo record with only a title. **Which single change matters most?**

- A. Add collaborators
 - B. Descriptive topics, matching keywords, and a description
 - C. More frequent commits
 - D. A shorter name
- B.** Discovery runs on *description*, not activity. Same reason a catalog record with no subject headings is nearly invisible.

Key points

- Description is what makes software findable, on GitHub, Zenodo, and beyond
- Keep it consistent across CITATION.cff, README, topics, and Zenodo: that's authority control